

Marc Berthelette

Software Engineer · Cloud Infrastructure · Python Platforms · Reliability

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linkedin.com/in/marc-berthelette-3aba6917 · Bilingual: French (native) · English (fluent) · Available immediately

Professional Summary

Software and cloud-infrastructure engineer with 18 years spanning production application development, internal platforms, DevOps, and reliability engineering. Built a Python/FastAPI validation service for 250 enterprise artifact repositories, wrote production software for roughly seven years, standardized CI/CD across 70+ applications, and automated Azure infrastructure with Terraform and Ansible. Takes ambiguous services from problem discovery through implementation and production operation, pairing software-engineering discipline with observability, incident response, documentation, and infrastructure ownership.

Core Expertise

Software Engineering	Python, FastAPI, C++, Java, Bash, SQL, NoSQL, API and service design, configurable validation systems
Cloud Infrastructure	Microsoft Azure, AWS, Terraform, Ansible, Puppet, Linux, environment provisioning, security and reliability
Containers & Delivery	Kubernetes, Docker, ArgoCD, Azure DevOps, GitHub Actions, Jenkins, CI/CD, pipeline-as-code
Production Reliability	Dynatrace, Splunk, ELK, incident response, RCA, automated remediation, monitoring and alerting
Engineering Enablement	Internal platforms, developer tooling, documentation, technical evaluation, cross-team adoption

Work Experience

Énergir — Senior DevOps Engineer

Montréal, QC | September 2021 – Present

- Owned CI/CD standardization for 12 application teams and 70+ applications across nine distinct runtimes, replacing manual delivery with automated pipelines and quality gates.
- Led cloud modernization for 5–10 mission-critical services on Microsoft Azure and implemented Terraform and Ansible infrastructure automation across development, staging, and production.
- Rebuilt Salesforce delivery with the official CLI and designed a validation-sandbox merge gate, reducing a recurring missed-deployment problem to one miss since launch.
- Drove incident RCA and implemented monitoring, observability dashboards, and automated remediation to reduce MTTR.
- Improved pipeline usability by separating application-code errors from platform errors in reporting, making failures self-diagnosing for engineering teams.
- Evaluated and introduced AI-assisted engineering practices with hard scope gates and human-validation checkpoints before generated code can reach production.

Key technologies: Python, Java, Microsoft Azure, Terraform, Ansible, Azure DevOps, GitHub Actions, CloudFoundry, Databricks, Salesforce

Desjardins — Senior DevOps Engineer (Contract)

Montréal, QC (Remote) | October 2025 – March 2026 — *concurrent engagement, delivered alongside the Énergir role*

- Designed and built a Python/FastAPI service that validated the configuration of roughly 250 enterprise JFrog repositories across seven package ecosystems every four hours and published results to Dynatrace.
- Made the validation engine tool-agnostic and its test set configurable, allowing administrators to extend it and turning the same service into the regression-safety net for the enterprise JFrog upgrade.
- Automated provisioning of on-demand Artifactory environments with Kubernetes and ArgoCD, providing self-service infrastructure for controlled testing.

- Implemented artifact-lifecycle policies and Splunk alerting for configuration changes made outside the CI/CD pipeline.

Key technologies: Python, FastAPI, Kubernetes, ArgoCD, GitHub Actions, JFrog Artifactory, Dynatrace, Splunk, Docker, Helm

Tink — DevOps Engineer

Montréal, QC | January 2018 – August 2021

- Conceived and built an internal software-inventory platform cataloguing every installed application and version across 600+ servers, exposing CVE, end-of-life, and SSL/TLS risks to product owners.
- Centralized Jenkins Pipeline as Code for 10–15 developers maintaining 80+ applications for 20+ clients.
- Automated Linux operating-system patching and kernel-update reboots across the fleet and managed configuration declaratively through Puppet and Spacewalk.
- Built ELK observability and a self-healing response for a recurring Elasticsearch failure, eliminating a class of manual on-call intervention.
- Built the compute, caching, database, DNS, and network layers of an AWS migration target; the client cancelled before production cutover.

Key technologies: Python, Bash, Linux, Jenkins, Puppet, Spacewalk, ELK Stack, AWS (EC2, S3, Route 53, VPC), VMware

SOVO Technologies — Lead Technician → Software Engineer

Montréal, QC | November 2008 – December 2017

- Designed and programmed the production scheduling system used for the production department's full planning cycle.
- Built a voice-recognition evaluation system and multilingual dictionary infrastructure supporting internationalized locales.
- Wrote production software for roughly seven years before the Software Engineer title formalized the work.
- Owned company infrastructure, managed a technician team, and served as the final quality gate before releases reached production.

Key technologies: C++, Java, SQL, Windows Server, Linux, Active Directory, VMware vSphere, Cisco routing

Education

Bachelor of Automated Manufacturing Engineering — École de Technologie Supérieure (ÉTS), Montréal, QC | 2005 – 2016

DCS in Computer Science Technology — Cégep André-Laurendeau, Montréal, QC | 2002 – 2005